

Solution Requirements

Date	03 October 2026
Team ID	SWTID-2026-6945
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	4 Marks

Functional Requirements (FR) define what ComicCraft does; Non-Functional Requirements (NFR) define how well it does it.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	No login is required in this release (single-session tool). API keys for Gemini and Hugging Face are kept server-side in a .env file and are never shown to users.	Low
2	Authorization levels	Single public user role: can create, preview and download comics. Developer-only utility route /test-image is intended for local testing.	Low
3	External interfaces	Google Gemini API via the google-genai SDK (Flash for outline, Pro for story); Hugging Face Hub / Diffusers to load Stable Diffusion; browser UI built with HTML, CSS and Jinja2.	High
4	Transactions processing	Each request runs the workflow: validate input, generate outline, story and five images, build layout, export PDF. Failures return clear messages (HTTP 422 / 500 / 502).	High
5	Reporting	On-screen comic preview (title, image, scene description, caption, narration) and downloadable timestamped PDF; JSON layout output via /generate-comic/json.	High
6	Business rules	Exactly 5 panels per comic; prompt 5-500 characters; setting, tone and art style come from fixed options; PDF names follow comic_YYYYMMDD_HHMMSS.pdf.	High
7	Compliance to laws or regulations	Follows Google Gemini and Stable Diffusion (CreativeML OpenRAIL-M) usage terms; no personal data is collected or stored; users should not enter personal or sensitive information.	Medium
8	Other	Offline mock mode (COMICCRAFT MOCK=1) for demos and tests; placeholder panels when the image model cannot load; interactive API docs at /docs.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Web pages respond instantly; comic generation time is dominated by the AI models. Use a GPU for image generation.	Static pages < 2 s; app overhead (excluding AI calls) < 2 s; full comic in minutes on a T4 GPU
2	Scalability	Stateless FastAPI app; heavy work runs in worker threads; can be run with several Uvicorn workers or containers.	Handles 10 concurrent users on app-level load without errors (mock AI)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
3	Security & Data Privacy	Secrets in .env (git-ignored); Pydantic input validation; download route restricted to comic_*.pdf files (path-traversal safe); no user data stored beyond generated files.	No secrets in repo; invalid names rejected with 400
4	Reliability & Availability	Retries with back-off on Gemini errors; graceful error pages instead of stack traces; placeholder image fallback.	0% errors in test runs; failures reported in < 1 s
5	Usability & Accessibility	Simple one-page form, labelled inputs, responsive layout, clear loading message, alt text on panel images.	New user can create a comic in under 1 minute of input
6	Other (Maintainability, Portability)	Modular services, docstrings, pytest suite, Dockerfile, configurable model IDs.	21 automated tests pass; runs on Windows, macOS and Linux